Location, Time, and Observer Based Three-Valued Logic: A Minimal Formal System Where Truth is Never Absolute

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Abstract

We present **LTO-K3**, a minimal formal system in which the truth value of a proposition is never absolute but is always evaluated with respect to three explicit parameters: a *location*, a *time*, and an *observer*. The system admits exactly three truth values—**T** (true), **F** (false), and **U** (unknown)—composed via Kleene's strong three-valued connectives. We give a formal syntax, a compositional point-wise semantics $\widehat{V} : \mathcal{L} \times L \times \Theta \times O \to \{\mathbf{T}, \mathbf{F}, \mathbf{U}\}$, two consequence relations (local and universal), and prove a small body of results: uniqueness of the extended valuation, locality, information-monotonicity, conservativity over classical logic, replacement of equivalents, the De Morgan laws, the absence of **T**-tautologies, and a reduction theorem that shows LTO-K3 at a fixed point is exactly Kleene's K_3. From the reduction theorem we inherit decidability, the co-NP-completeness of consequence, and the existence of sound and complete proof systems. We are careful to distinguish what is genuinely new (the *packaging*: location and observer as non-optional, concrete parameters; a deliberate refusal to introduce modal operators) from what is inherited from established work [2, 1, 3, 4, 5]. The contribution is therefore a careful synthesis with explicit, proven properties, not a discovery, and we name its limitations explicitly.

Keywords: three-valued logic, Kleene logic, indexed semantics, temporal logic, epistemic logic, contextual semantics, partial information.

1 Introduction

Classical two-valued logic forces every proposition into one of two states: *true* or *false*. In many practical settings—verification of distributed systems, knowledge representation, robotics, legal reasoning, data quality—this dichotomy is too coarse. Truth often depends on *where* an evaluation is performed, *when* it is performed, and *who* performs it, and in many cases the evaluator simply does not know.

We introduce a minimal logic, **LTO-K3** (Location, Time, Observer; Kleene three-valued), that takes these dependencies as primitive. Every atomic proposition is evaluated under an explicit tuple (l, θ, o) of location, time, and observer, and takes one of three values: true, false, or unknown. Compound formulas are evaluated point-wise via Kleene's strong three-valued tables.

The system is intentionally austere: no modal operators, no accessibility relations, no transition systems, no quantification over locations, times, or observers. Its purpose is to serve as a compact, unambiguous core on which richer domain-specific layers may be built.

Contributions.

1. A formal syntax and compositional point-wise semantics for a three-valued indexed logic over (L, Θ, O) frames (Sections 3–4).

2. A *reduction theorem* (Theorem 8) showing that at every fixed point (l, θ, o), LTO-K3 is exactly Kleene's K_3, whence all meta-theoretic results for K_3 lift to LTO-K3.

3. Two consequence relations—local at a point and universal across models and points—together with an explicit example separating them (Section 6).
4. A small body of meta-theoretic results (Section 7): uniqueness, locality, information-monotonicity, conservativity over classical logic, replacement of equivalents, De Morgan laws, absence of **T**-tautologies.
5. Decidability and complexity results (Section 8): model checking is linear; local and universal consequence are co-NP-complete; sound and complete proof systems exist (inherited from K_3).
6. A worked example (Section 9) and an explicit comparison with the closest neighbours: Kleene K_3, Łukasiewicz Ł$_3$, Bochvar B_3, Priest's LP, Belnap's **FOUR**, hybrid logic, and multi-agent epistemic temporal logic (Section 10).
7. A frank limitations section (Section 11).

What is and is not new. None of the parts is new. Three-valued logics go back to Łukasiewicz [1] and Kleene [2]. Temporal indexing goes back to Prior [3]; possible-worlds indexing to Kripke [5]; agent indexing to Hintikka [4] and, in computer science, to Fagin et al. [12]. Sound and complete proof systems for K_3 (tableaux, signed-clause resolution) are well known [11]. What is new is the *packaging*: location, time, and observer are simultaneously non-optional, concrete, first-class parameters of a single valuation function, and we deliberately refuse to lift them with modal operators. The reduction theorem makes precise the sense in which our contribution is a packaging and not a new logic.

2 Preliminaries: Kleene's Strong Three-Valued Logic

Kleene's strong three-valued logic K_3 [2] extends classical two-valued logic with a third value **U** ("unknown" / "undefined") and defines the connectives so that **U** propagates conservatively: a connective returns **T** (resp. **F**) iff every way of resolving each **U** to **T** or **F** would yield **T** (resp. **F**); otherwise it returns **U**. The truth tables are:

$\wedge$	**T**	**U**	**F**
T	**T**	**U**	**F**
U	**U**	**U**	**F**
F	**F**	**F**	**F**

$\vee$	**T**	**U**	**F**
T	**T**	**T**	**T**
U	**T**	**U**	**U**
F	**T**	**U**	**F**

p	$\neg p$
T	**F**
U	**U**
F	**T**

We treat material implication as a defined abbreviation: $\varphi \to \psi \;:=\; \neg\varphi \vee \psi$. This is strong-Kleene implication; Łukasiewicz's Ł$_3$ uses a different connective, where $\mathbf{U} \to_{Ł_3} \mathbf{U} = \mathbf{T}$ rather than **U** (see Section 10). The induced truth table for $\to$ is:

$\to$	**T**	**U**	**F**
T	**T**	**U**	**F**
U	**T**	**U**	**U**
F	**T**	**T**	**T**

Information ordering. Define the partial order $\sqsubseteq$ on $\{\mathbf{T}, \mathbf{F}, \mathbf{U}\}$ by $\mathbf{U} \sqsubseteq \mathbf{T}$, $\mathbf{U} \sqsubseteq \mathbf{F}$, and $x \sqsubseteq x$ for all x; **T** and **F** are incomparable. Intuitively, $x \sqsubseteq y$ means "y contains at least as much information as x." All Kleene connectives are *monotone* in $\sqsubseteq$ component-wise: resolving an **U** to **T** or **F** never reverses an already-determined value. This is the formal reason K_3 is appropriate when **U** denotes epistemic uncertainty.

Functional incompleteness. K_3 is not functionally complete on $\{\mathbf{T}, \mathbf{F}, \mathbf{U}\}$: every formula built from $\{\neg, \wedge, \vee\}$ over atoms valued $\mathbf{U}$ evaluates to $\mathbf{U}$ (immediate from the truth tables). In particular there are no constants $\mathbf{T}, \mathbf{F}$ definable from the connectives alone. This fact underlies the absence of $\mathbf{T}$-tautologies (Theorem 21).

3 Syntax

Fix a countably infinite set AP of *atomic propositions* $p, q, r, \ldots$. The language $\mathcal{L}$ is generated by the grammar

$$\varphi, \psi \ ::= \ p \ | \ \neg\varphi \ | \ (\varphi \wedge \psi) \ | \ (\varphi \vee \psi), \qquad p \in \mathrm{AP}.$$

We omit parentheses under the usual precedence $\neg > \wedge > \vee$, and use $\varphi \to \psi$ as an abbreviation for $\neg\varphi \vee \psi$. The set of atomic propositions occurring in φ is written $\mathrm{AP}(\varphi)$.

Remark 1. The biconditional $\leftrightarrow$ and the quantifiers $\forall, \exists$ are deliberately omitted. The biconditional could be defined as $(\varphi \to \psi) \wedge (\psi \to \varphi)$ but its K_3 behaviour is non-standard and we prefer to omit it. Quantifiers would force a domain commitment orthogonal to indexing. Crucially, $\mathcal{L}$ contains *no modal operators* indexed over L, Θ, O (no $\Box_l$, no $\mathsf{F}\theta$, no $K_o\varphi$). This is the essential design choice: indices appear in the meta-language, not in the object language.

4 Semantics

4.1 Frames and models

Definition 2 (Frame)**.** A *frame* is a triple $\mathcal{F} = (L, \Theta, O)$ where L, Θ, and O are non-empty sets of *locations*, *times*, and *observers*, respectively. We call $(l, \theta, o) \in L \times \Theta \times O$ a *point* of the frame.

We impose no structure on L, Θ, or O: Θ is not required to be ordered, L not required to carry a topology, O not required to carry an information-state relation. Any such structure is a domain-specific extension; see Section 11.

Definition 3 (Model)**.** An *LTO-K3 model* is a pair $\mathcal{M} = (\mathcal{F}, V)$ where $\mathcal{F} = (L, \Theta, O)$ is a frame and

$$V : \mathrm{AP} \times L \times \Theta \times O \ \longrightarrow \ \{\mathbf{T}, \mathbf{F}, \mathbf{U}\}$$

is a total *base valuation* assigning a truth value to each atomic proposition at each point.

4.2 Compositional valuation

Definition 4 (Extended valuation)**.** Let $\mathcal{M} = (\mathcal{F}, V)$ be a model. The *extended valuation* $\widehat{V} : \mathcal{L} \times L \times \Theta \times O \to \{\mathbf{T}, \mathbf{F}, \mathbf{U}\}$ is defined by structural recursion on φ:

$$\begin{aligned}
\widehat{V}(p, l, \theta, o) &= V(p, l, \theta, o), \quad p \in \mathrm{AP}, \\
\widehat{V}(\neg\varphi, l, \theta, o) &= \neg_{K_3}\widehat{V}(\varphi, l, \theta, o), \\
\widehat{V}(\varphi \wedge \psi, l, \theta, o) &= \widehat{V}(\varphi, l, \theta, o) \wedge_{K_3} \widehat{V}(\psi, l, \theta, o), \\
\widehat{V}(\varphi \vee \psi, l, \theta, o) &= \widehat{V}(\varphi, l, \theta, o) \vee_{K_3} \widehat{V}(\psi, l, \theta, o),
\end{aligned}$$

where $\neg_{K_3}, \wedge_{K_3}, \vee_{K_3}$ are the Kleene connectives of Section 2.

Proposition 5 (Existence and uniqueness of $\widehat{V}$)**.** *For every model $\mathcal{M}$, the function $\widehat{V}$ defined in Definition 4 exists and is unique.*

Proof. $\mathcal{L}$ is the absolutely free algebra on AP over the signature $\{\neg, \wedge, \vee\}$ (with arities $1, 2, 2$). At each fixed point (l, θ, o), $(\{\mathbf{T}, \mathbf{F}, \mathbf{U}\}, \neg_{K_3}, \wedge_{K_3}, \vee_{K_3})$ is an algebra of the same signature. The map $p \mapsto V(p, l, \theta, o)$ on generators extends, by the universal property of free algebras, to a unique homomorphism $\widehat{V}(-, l, \theta, o) : \mathcal{L} \to \{\mathbf{T}, \mathbf{F}, \mathbf{U}\}$. Doing this at every point yields $\widehat{V}$. $\square$

Crucially, the recursion is *point-wise*: both sides of every binary connective are evaluated at *the same* point. There is no syntactic mechanism in $\mathcal{L}$ for referring to a different point.

Remark 6 (On cross-frame composition). $\mathcal{L}$ contains no formula whose two sub-formulas can be evaluated at different points; Definition 4 forces a single (l, θ, o) throughout. Defining cross-frame operators would require explicit bridging rules (synchronization, trust, observer alignment) that are domain-specific. Such extensions are outside the core system; see Section 11.

5 Reduction to Pointwise K_3

This section formalises the sense in which LTO-K3 is, at every fixed point, exactly Kleene's K_3. The reduction theorem is the central meta-fact of the paper: it lets us inherit completeness, decidability, and proof systems from K_3 without re-proving them.

Definition 7 (Point restriction)**.** Let $\mathcal{M} = (\mathcal{F}, V)$ be a model and (l, θ, o) a point. The K_3-*assignment induced at* (l, θ, o) is the function $v^{\mathcal{M}}_{(l,\theta,o)} : \mathrm{AP} \to \{\mathbf{T}, \mathbf{F}, \mathbf{U}\}$ defined by $v^{\mathcal{M}}_{(l,\theta,o)}(p) = V(p, l, \theta, o)$.

Let $\widehat{V}_{K_3} : \mathcal{L} \times (\mathrm{AP} \to \{\mathbf{T}, \mathbf{F}, \mathbf{U}\}) \to \{\mathbf{T}, \mathbf{F}, \mathbf{U}\}$ denote the standard K_3 semantics: $\widehat{V}_{K_3}(\varphi, v)$ is the value of φ under the assignment v using the Kleene tables.

Theorem 8 (Reduction)**.** *For every model $\mathcal{M}$, every formula $\varphi \in \mathcal{L}$, and every point (l, θ, o),*

$$\widehat{V}(\varphi, l, \theta, o) \;=\; \widehat{V}_{K_3}(\varphi,\, v^{\mathcal{M}}_{(l,\theta,o)}).$$

Proof. Structural induction on φ. Base: for $p \in \mathrm{AP}$ both sides equal $V(p, l, \theta, o)$. Step: each compound case applies exactly the same Kleene connective on either side, and the induction hypothesis equates the sub-formula values. □

Corollary 9 (Inheritance from K_3)**.** *Every property of K_3 formulated in terms of a single assignment lifts point-wise to LTO-K3. In particular, the propositional fragment of LTO-K3 inherits from K_3:*

1. *the absence of* $\mathbf{T}$*-tautologies (Theorem 21 below),*
2. *the existence of sound and complete proof systems (tableaux, signed-clause resolution) [11],*
3. *the co-NP-completeness of consequence (Section 8).*

6 Consequence Relations

We take $\mathbf{T}$ as the sole designated value. (Designating $\{\mathbf{T}, \mathbf{U}\}$ gives Priest's LP [8], a different logic.)

Definition 10 (Local satisfaction)**.** Let $\mathcal{M} = (\mathcal{F}, V)$ be a model and (l, θ, o) a point. We say φ is *satisfied at* (l, θ, o) *in* $\mathcal{M}$, written $\mathcal{M}, (l, \theta, o) \vDash \varphi$, iff $\widehat{V}(\varphi, l, \theta, o) = \mathbf{T}$.

Definition 11 (Consequence)**.** Let $\Gamma \subseteq \mathcal{L}$ and $\varphi \in \mathcal{L}$.

- *Local entailment at* (l, θ, o) *in* $\mathcal{M}$, written $\Gamma \vDash_{\mathcal{M},(l,\theta,o)} \varphi$: if every $\gamma \in \Gamma$ satisfies $\mathcal{M}, (l, \theta, o) \vDash \gamma$, then $\mathcal{M}, (l, \theta, o) \vDash \varphi$.
- *Universal entailment* $\Gamma \vDash \varphi$: $\Gamma \vDash_{\mathcal{M},(l,\theta,o)} \varphi$ for every model $\mathcal{M}$ and every point (l, θ, o) of $\mathcal{M}$.

Proposition 12 (Universal entails local)**.** $\Gamma \vDash \varphi$ *implies* $\Gamma \vDash_{\mathcal{M},(l,\theta,o)} \varphi$ *for every $\mathcal{M}$ and (l, θ, o). The converse fails.*

Proof. The forward direction is immediate from the definition. For failure of the converse, take any frame with two points $(l,\theta,o)_1 \neq (l,\theta,o)_2$, an atom $p \in \mathrm{AP}$, and the model V defined by $V(p,(l,\theta,o)_1) = \mathbf{T}$, $V(p,(l,\theta,o)_2) = \mathbf{F}$, and $V(q,\cdot) = \mathbf{U}$ for every other atom q and every point. Then $\varnothing \vDash_{\mathcal{M},(l,\theta,o)_1} p$ (since $\mathcal{M},(l,\theta,o)_1 \vDash p$) but $\varnothing \nvDash p$ (Theorem 21). □

Remark 13. By Theorem 8, universal entailment coincides with K_3-consequence at the level of formulas: $\Gamma \vDash \varphi$ iff for every K_3-assignment $v : \mathrm{AP} \to \{\mathbf{T},\mathbf{F},\mathbf{U}\}$, if $\widehat{V}_{K_3}(\gamma, v) = \mathbf{T}$ for every $\gamma \in \Gamma$ then $\widehat{V}_{K_3}(\varphi, v) = \mathbf{T}$. Indexing adds nothing at the universal level; it matters only locally.

7 Properties

We collect the basic meta-theoretic properties of LTO-K3. Most are immediate corollaries of Theorem 8; we state them for completeness and to fix notation for the indexed setting.

Lemma 14 (Locality)**.** *For every model $\mathcal{M}$, every formula φ, and every point (l,θ,o), $\widehat{V}(\varphi,l,\theta,o)$ depends only on the restriction of V to $\mathrm{AP}(\varphi) \times \{l\} \times \{\theta\} \times \{o\}$.*

Proof. By Theorem 8, $\widehat{V}(\varphi,l,\theta,o) = \widehat{V}_{K_3}(\varphi, v^{\mathcal{M}}_{(l,\theta,o)})$, and standard K_3-semantics depends only on the values of $v^{\mathcal{M}}_{(l,\theta,o)}$ on $\mathrm{AP}(\varphi)$. □

Proposition 15 (Information-monotonicity)**.** *Let V_1, V_2 be base valuations on the same frame with $V_1(p,l,\theta,o) \sqsubseteq V_2(p,l,\theta,o)$ for every atom and every point. Then for every formula φ and every point (l,θ,o), $\widehat{V}_1(\varphi,l,\theta,o) \sqsubseteq \widehat{V}_2(\varphi,l,\theta,o)$.*

Proof. At a fixed (l,θ,o), both sides equal the K_3-evaluation of φ under the induced assignments, and $\neg_{K_3}, \wedge_{K_3}, \vee_{K_3}$ are monotone in $\sqsubseteq$ component-wise (standard K_3 result, verified by inspection of the truth tables). Structural induction completes the proof. □

Corollary 16 (Conservativity over classical logic)**.** *Let $\mathcal{M}$ be a model in which V never takes the value $\mathbf{U}$. Then for every formula φ and every point (l,θ,o), $\widehat{V}(\varphi,l,\theta,o) \in \{\mathbf{T},\mathbf{F}\}$, and the resulting value coincides with the classical (two-valued) valuation of φ under the same atomic assignment.*

Proof. When neither input is $\mathbf{U}$, every Kleene connective returns the classical value (inspection). The corollary follows by structural induction (alternatively, by Theorem 8 and the analogous K_3 result). □

Notation 17 (Semantic equivalence)**.** Write $\varphi \equiv \psi$ iff $\widehat{V}(\varphi,l,\theta,o) = \widehat{V}(\psi,l,\theta,o)$ for every model $\mathcal{M}$ and every point (l,θ,o).

Lemma 18 (Replacement of equivalents)**.** *If $\varphi \equiv \psi$ then for every formula χ and every occurrence of φ in χ, replacing that occurrence by ψ yields a formula χ' with $\chi \equiv \chi'$.*

Proof. Structural induction on χ. The base case ($\chi = \varphi$) is by hypothesis. The compound cases follow because each Kleene connective is a function of the truth values of its sub-formulas: replacing a sub-formula by an equivalent one does not change those values, so the top-level value is unchanged. (Equivalently: $\equiv$ is a congruence on the formula algebra because Kleene connectives are well-defined functions on $\{\mathbf{T},\mathbf{F},\mathbf{U}\}$.) □

Proposition 19 (De Morgan laws)**.** *For all φ,ψ: $\neg(\varphi \wedge \psi) \equiv \neg\varphi \vee \neg\psi$ and $\neg(\varphi \vee \psi) \equiv \neg\varphi \wedge \neg\psi$.*

Proof. At a fixed point, by case analysis on $(\widehat{V}(\varphi), \widehat{V}(\psi)) \in \{\mathbf{T},\mathbf{F},\mathbf{U}\} \times \{\mathbf{T},\mathbf{F},\mathbf{U}\}$ (nine cases per identity) using the Kleene tables. □

Proposition 20 (Standard identities)**.** *For all* φ, ψ, χ*:* $\neg\neg\varphi \equiv \varphi$*;* $\varphi \wedge \varphi \equiv \varphi$*;* $\varphi \vee \varphi \equiv \varphi$*;* $\varphi \wedge \psi \equiv \psi \wedge \varphi$*;* $\varphi \vee \psi \equiv \psi \vee \varphi$*;* $(\varphi \wedge \psi) \wedge \chi \equiv \varphi \wedge (\psi \wedge \chi)$*;* $(\varphi \vee \psi) \vee \chi \equiv \varphi \vee (\psi \vee \chi)$*;* $\varphi \wedge (\psi \vee \chi) \equiv (\varphi \wedge \psi) \vee (\varphi \wedge \chi)$*;* $\varphi \vee (\psi \wedge \chi) \equiv (\varphi \vee \psi) \wedge (\varphi \vee \chi)$*.*

Proof. Each is a finite truth-table check on $\{\mathbf{T}, \mathbf{F}, \mathbf{U}\}$, lifted point-wise via Theorem 8. □

Theorem 21 (No universal **T**-tautologies)**.** *There is no formula* $\varphi \in \mathcal{L}$ *with* $\vDash \varphi$*.*

Proof. Consider the model $\mathcal{M}_{\mathbf{U}}$ on any frame with $V(p,(l,\theta,o)) = \mathbf{U}$ for every atom and every point. By structural induction on φ, using that every Kleene connective returns **U** when every input is **U**, we have $\widehat{V}(\varphi,(l,\theta,o)) = \mathbf{U}$ for every φ and every (l,θ,o). Hence no formula evaluates to **T** universally. □

Corollary 22. $\nvDash p \vee \neg p$ *and* $\nvDash \neg(p \wedge \neg p)$*.*

Remark 23. Theorem 21 is the price of taking **T** as the sole designated value in K_3. It does not mean the logic is useless: valid *inferences* from non-empty contexts are the appropriate notion of validity. For example, $\{p, \neg p \vee q\} \vDash q$ holds: at any point where $\widehat{V}(p) = \mathbf{T}$, the table for $\vee$ forces $\widehat{V}(\neg p \vee q) = \widehat{V}(q)$, so if the latter is **T** then so is the former.

8 Decidability and Complexity

We assume formulas, finite subsets of $\Gamma \subseteq \mathcal{L}$, finite fragments of frames, and base valuations to be presented in the usual finitary way (formulas as syntax trees; valuations as finite tables on the atoms and points actually mentioned). Sizes refer to the total description length.

Proposition 24 (Model checking)**.** *Given a finite model* $\mathcal{M}$ *(i.e., a finite table for* V *on the points and atoms used) and a formula* φ*, computing* $\widehat{V}(\varphi,(l,\theta,o))$ *takes time* $O(|\varphi|)$ *assuming constant-time lookup of* V *at the relevant atoms.*

Proof. Bottom-up evaluation of the syntax tree of φ, applying a constant-time Kleene table lookup at each internal node. □

Theorem 25 (Complexity of consequence)**.** *Universal entailment* $\Gamma \vDash \varphi$ *for finite* Γ *is co-NP-complete. So is local entailment for finite* Γ*.*

Proof. By Theorem 8, universal entailment coincides with K_3-consequence at the level of formulas, which is known to be co-NP-complete (it reduces to classical SAT/VAL via the standard embedding of K_3 into classical two-valued logic; see, e.g., [11]). Local entailment at a specified point is at least as hard (universal entailment is the intersection over points), and is in co-NP by guessing a falsifying assignment at the point and checking in linear time (Proposition 24). Membership and hardness match, giving co-NP-completeness. □

Corollary 26 (Sound and complete proof systems)**.** *LTO-K3 inherits, via Theorem 8, the sound and complete tableau and signed-clause-resolution proof systems for* K_3 *[11]. A formal derivation of* $\Gamma \vDash \varphi$ *in LTO-K3 is a* K_3*-derivation of* φ *from* Γ *at the level of formulas.*

9 Worked Example

Let $L = \{l_{\text{EU}}, l_{\text{US}}\}$, $\Theta = \{\theta_0, \theta_1\}$, $O = \{o_{\text{Alice}}, o_{\text{Bob}}\}$. Let p abbreviate "the package has been delivered" and q "the package has cleared customs." A base valuation:

l	θ	o	$V(p)$	$V(q)$
l_{EU}	θ_0	o_{Alice}	**U**	**F**
l_{EU}	θ_1	o_{Alice}	**T**	**T**
l_{EU}	θ_1	o_{Bob}	**U**	**T**
l_{US}	θ_1	o_{Alice}	**F**	**U**

Then $\widehat{V}(p \wedge q)$ takes value $\mathbf{T}$ at $(l_{\text{EU}}, \theta_1, o_{\text{Alice}})$ (Alice in the EU at θ_1 knows both); $\mathbf{U}$ at $(l_{\text{EU}}, \theta_1, o_{\text{Bob}})$ (Bob does not yet know whether the package has been delivered, even though he knows it has cleared customs); and $\mathbf{F}$ at $(l_{\text{US}}, \theta_1, o_{\text{Alice}})$ (the proposition is false in the US frame because $\widehat{V}(p) = \mathbf{F}$ there). The same formula yields different values at different points: this is the design.

By Theorem 8, each row of the table determines a K_3-assignment, and the value of $p \wedge q$ at that row is exactly the K_3-evaluation of $p \wedge q$ under that assignment.

10 Related Work

Three-valued logics. K_3 [2] is one of several three-valued logics with different motivations and connectives. Łukasiewicz Ł$_3$ [1] differs from K_3 only in implication: $\mathbf{U} \to_{Ł_3} \mathbf{U} = \mathbf{T}$, yielding the tautology $p \to p$ and other classical theorems. Bochvar's internal B_3 [9] is "infectious": any operation with a $\mathbf{U}$ input returns $\mathbf{U}$, even $\mathbf{T} \vee \mathbf{U}$. Priest's LP [8] uses the Kleene tables but designates $\{\mathbf{T}, \mathbf{U}\}$, yielding a paraconsistent logic in which contradictions can be true and LEM holds. Belnap's **FOUR** [6] adds a fourth value ("both true and false") for tolerating contradictory information.

We choose K_3 with designated value $\mathbf{T}$ because the reading of $\mathbf{U}$ we want is purely epistemic ("not yet known") and information-monotonic (Proposition 15). Ł$_3$ breaks this property (implication is not monotone in the obvious sense); LP changes the designated set; B_3 propagates $\mathbf{U}$ too aggressively; **FOUR** conflates uncertainty with inconsistency.

Temporal and modal logics. Prior [3] introduced tense operators; Pnueli [7] introduced LTL; Clarke and Emerson [10] introduced CTL. These logics index truth by time (or states reachable in time) but use classical two-valued semantics and quantify over states with modal operators. Kripke [5] gave possible-worlds semantics; the entire modal-logic tradition [13] indexes truth by worlds and uses $\Box, \Diamond$ to quantify over them. Hybrid logic [15] adds nominals i and $@_i$ to refer to specific worlds in the object language. LTO-K3 differs in not quantifying over points in the object language at all.

Epistemic logics. Hintikka [4] introduced epistemic modal logic with K_a operators; see Fagin et al. [12] for the textbook treatment. Multi-agent epistemic temporal logic combines these dimensions, but again with two-valued semantics and modal operators.

Three-valued spatio-temporal and runtime-verification logics. Bauer et al. [16] developed LTL$_3$, a three-valued LTL variant for runtime monitoring. Spatial logics [14] introduce explicit spatial indices for reasoning about topological structure. These works are the closest neighbours of LTO-K3.

Honest positioning. Relative to the above, LTO-K3 is deliberately less expressive: it has fewer resources than hybrid logic, fewer than multi-modal epistemic temporal logic, and fewer than LTL$_3$. What it offers in return is a uniform treatment of three indices (location, time, observer) as concrete and equally first-class, and a self-contained semantics that fits on a page. Theorem 8 makes precise the price of this minimalism: at any single point, LTO-K3 is exactly

K_3 and nothing more. We make no claim that LTO-K3 supersedes any of these systems; we claim only that it is a useful *starting point.*

11 Limitations

1. **No cross-frame operators.** LTO-K3 cannot express "φ at l_1 implies ψ at l_2" or "φ now will be ψ later."

2. **No quantification.** LTO-K3 cannot express "φ holds for some observer" or "φ holds at all times."

3. **Structure on L, Θ, O is invisible.** Time is not ordered, locations are not topologized, observers do not have information-state relations. Such structure must be added by extension.

4. **Functional incompleteness and no T-tautologies.** Inherited from K_3 (Section 2, Theorem 21). Users accustomed to tautological reasoning must adjust to entailment-based reasoning from non-empty contexts.

5. **U is undifferentiated.** "Not yet known," "in principle unknowable," and "not yet defined" collapse into one value. Distinguishing them requires a richer truth-value set [6] or a probabilistic / possibilistic layer.

6. **No interaction between observers.** There is no notion of "o_1 knows that o_2 knows φ"; that requires explicit K_o operators (out of scope).

7. **Cross-frame undefinedness is a restriction.** We have framed it as a deliberate design choice, and it is, but it is also a real expressive limitation. Users who need cross-frame reasoning will need to layer on additional machinery.

8. **No gain over K_3 at a fixed point.** By Theorem 8, the indexing contributes nothing at a single point. The value of LTO-K3 is therefore not deeper logic but uniform treatment of context across points and a discipline that forbids silent context-mixing.

12 Conclusion

We have presented LTO-K3, a minimal three-valued logic in which every proposition is evaluated under an explicit triple of location, time, and observer. We gave a formal syntax, a compositional point-wise semantics, two consequence relations, a reduction theorem that anchors LTO-K3 in Kleene's K_3, and a small body of meta-theoretic results (uniqueness, locality, monotonicity, conservativity, replacement, De Morgan, absence of **T**-tautologies). The reduction theorem lets us inherit decidability, the co-NP-completeness of consequence, and sound and complete proof systems from K_3. We were explicit about what is new (the packaging) and what is not (everything else), and named the system's limitations. Cross-frame composition, quantification, and modal operators are deliberate omissions to be addressed by domain-specific extensions.

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